Severe Weather Events:
Health and Economic Consequences

NOAA Storm Database analysis

Course project report

Records analyzed	902,297
Largest health impact	TORNADO - 96,979 fatalities + injuries
Largest economic impact	FLOOD - USD 150.32 billion
Source	Official assignment file: StormData.csv.bz2

This report presents an analysis of the supplied NOAA Storm Database.
The accompanying R Markdown file documents the data processing and analysis.

Synopsis

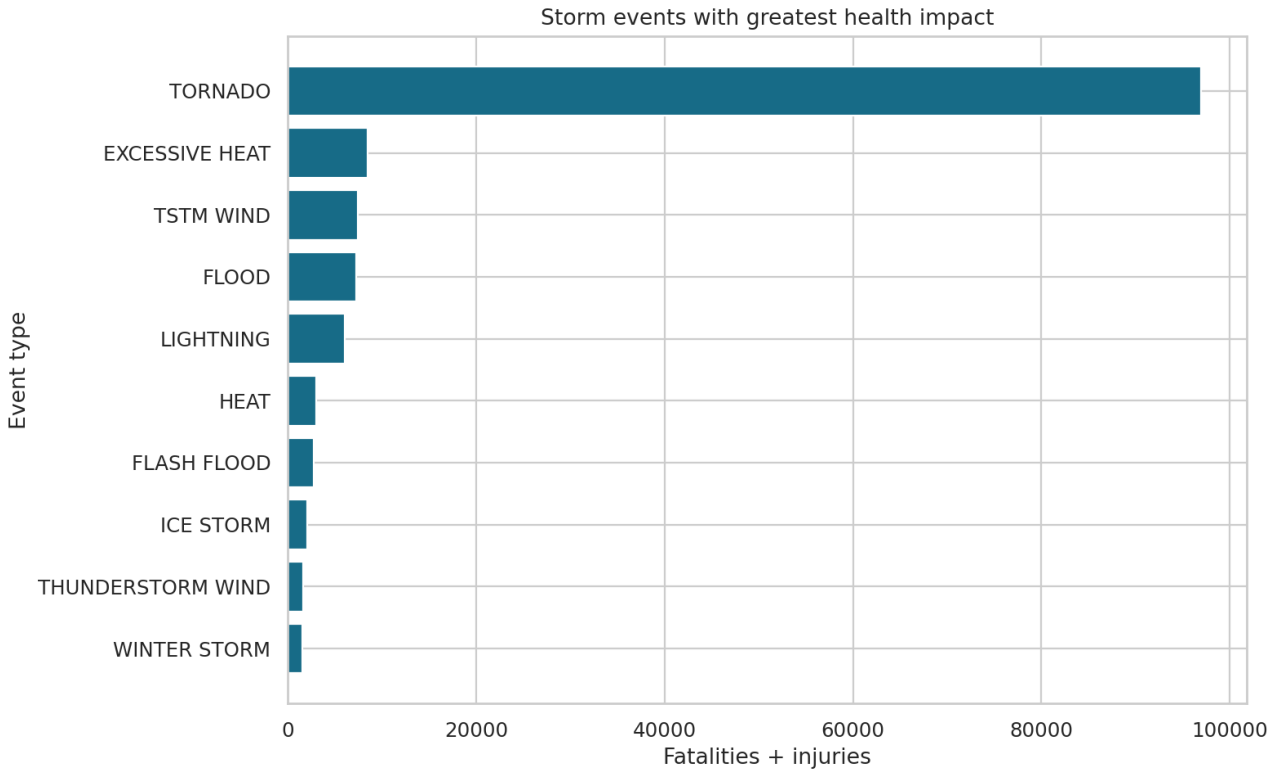
This analysis uses the full NOAA Storm Database supplied for the assignment to identify event types associated with the greatest public-health and economic consequences. Event labels are converted to uppercase and trimmed, but otherwise retained so that the transformation is transparent. Public-health impact is measured as fatalities plus injuries. Economic impact is reported property damage plus crop damage after interpreting magnitude fields. Tornadoes have the largest combined health impact, with 96,979 fatalities and injuries. Floods have the greatest estimated economic impact, approximately USD 150.32 billion, followed by hurricanes/typhoons and tornadoes. Historical inconsistencies in event names and magnitude codes limit direct comparison, so conclusions concern recorded impacts in this file rather than causal risk.

Data Processing

The analysis begins with StormData.csv.bz2 and retains the seven columns needed for the two questions. EVTYPE is uppercased and trimmed. Health impact equals FATALITIES + INJURIES. Damage exponents use H=10^2, K=10^3, M=10^6, B=10^9, and numeric codes 0 through 8 as powers of ten. Blank and other symbols receive multiplier one. The data contain 14 unrecognized property exponent rows and 7 unrecognized crop exponent rows; this choice is disclosed because exponent coding is historically inconsistent.

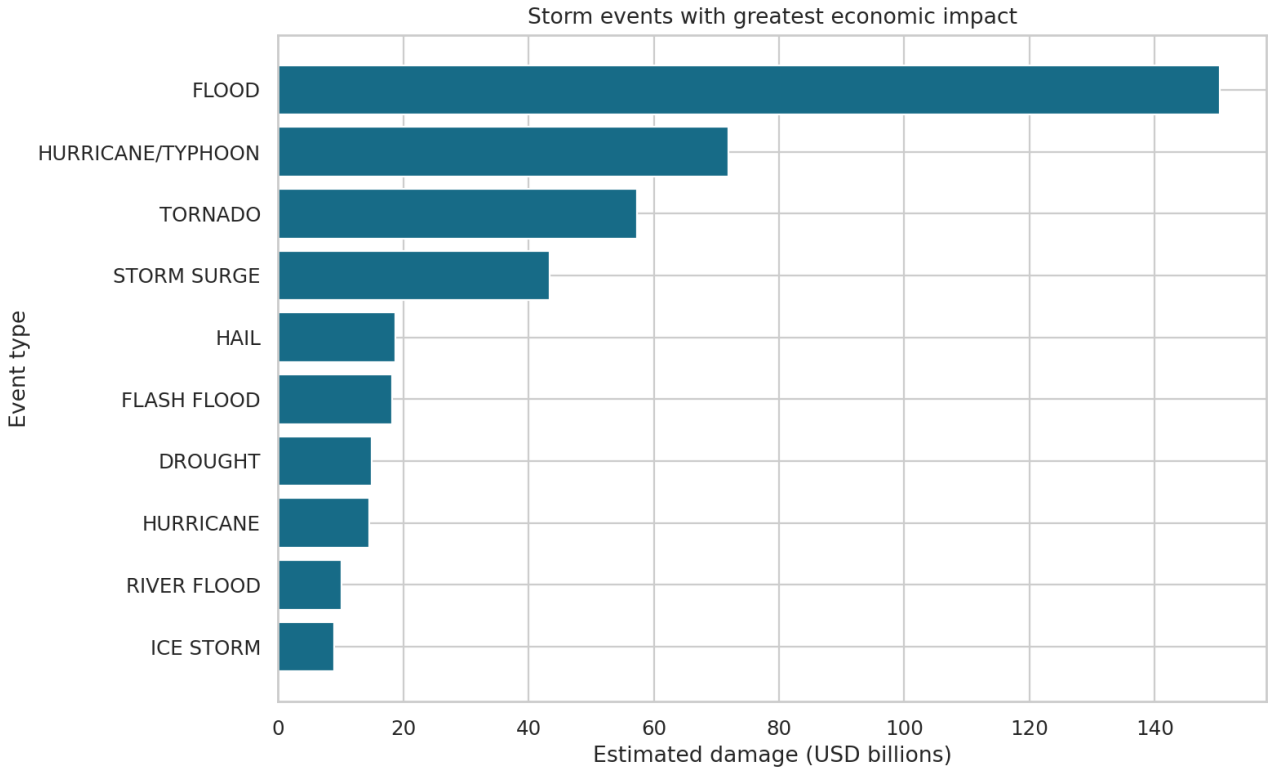
Results

Population health

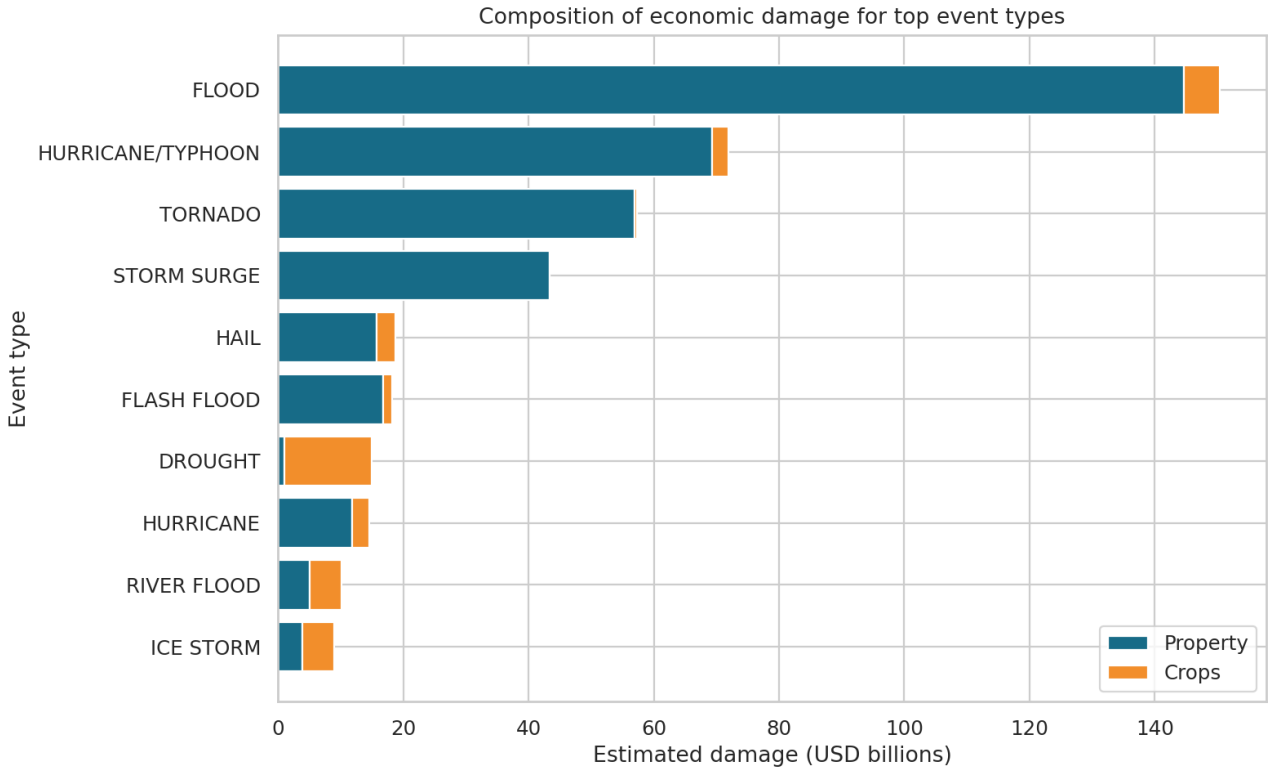


Tornadoes are the largest recorded health burden. Excessive heat and thunderstorm wind are the next highest categories. The combined metric does not imply that a fatality and an injury have equal severity.

Economic consequences



Floods produce the greatest estimated combined property and crop loss, about USD 150.32 billion. Hurricanes/typhoons rank second and tornadoes third. Values are nominal amounts as recorded; no inflation adjustment is applied.



Property damage dominates most top-ranked categories. Crop loss contributes materially to drought and flood totals.

Reproducible R code

The following code is the complete analysis code included in the companion R Markdown source.

Code block 1

```
storm <- read.csv("data/StormData.csv.bz2", stringsAsFactors = FALSE,
                  fileEncoding = "latin1")
keep <- c("EVTYPE", "FATALITIES", "INJURIES", "PROPDMG", "PROPDMGEXP",
          "CROPDMG", "CROPDMGEXP")
storm <- storm[keep]
storm$event <- toupper(trimws(storm$EVTYPE))
```

Code block 2

```
damage_multiplier <- function(x) {
  x <- toupper(trimws(ifelse(is.na(x), "", x)))
  out <- rep(1, length(x))
  out[x == "H"] <- 1e2
  out[x == "K"] <- 1e3
  out[x == "M"] <- 1e6
  out[x == "B"] <- 1e9
  for (n in 0:8) out[x == as.character(n)] <- 10^n
  out
}

storm$health <- ifelse(is.na(storm$FATALITIES), 0, storm$FATALITIES) +
                ifelse(is.na(storm$INJURIES), 0, storm$INJURIES)
storm$property_loss <- ifelse(is.na(storm$PROPDMG), 0, storm$PROPDMG) *
                       damage_multiplier(storm$PROPDMGEXP)
storm$crop_loss <- ifelse(is.na(storm$CROPDMG), 0, storm$CROPDMG) *
                   damage_multiplier(storm$CROPDMGEXP)
storm$economic_loss <- storm$property_loss + storm$crop_loss

health <- aggregate(cbind(FATALITIES, INJURIES, health) ~ event, storm, sum, na.rm=TRUE)
health <- health[order(health$health, decreasing=TRUE), ]
economic <- aggregate(cbind(property_loss, crop_loss, economic_loss) ~ event,
                      storm, sum, na.rm=TRUE)
economic <- economic[order(economic$economic_loss, decreasing=TRUE), ]
```

Code block 3

```
top.health <- health[1:10, ]
par(mar=c(5, 11, 3, 1))
barplot(rev(top.health$health), names.arg=rev(top.health$event), horiz=TRUE,
        las=1, col="#176B87", main="Storm events with greatest health impact",
        xlab="Fatalities + injuries")
```

Code block 4

```
top.economic <- economic[1:10, ]
par(mar=c(5, 11, 3, 1))
barplot(rev(top.economic$economic_loss) / 1e9,
        names.arg=rev(top.economic$event), horiz=TRUE, las=1,
        col="#176B87", main="Storm events with greatest economic impact",
        xlab="Estimated damage (USD billions)")
```

Code block 5

```
m <- rbind(rev(top.economic$property_loss), rev(top.economic$crop_loss)) / 1e9
par(mar=c(5, 11, 3, 1))
barplot(m, names.arg=rev(top.economic$event), horiz=TRUE, las=1,
        col=c("#176B87", "#F28E2B"), main="Composition of economic damage",
        xlab="Estimated damage (USD billions)", legend.text=c("Property", "Crops"))
```

Limitations

Event names are not fully standardized in the historical database, so related phenomena can appear under different labels. Magnitude codes include a small number of nonstandard symbols. Monetary values are not adjusted for inflation, and database coverage changes over time. The ranking therefore summarizes recorded consequences in the supplied file and should not be interpreted as a causal estimate of future risk.